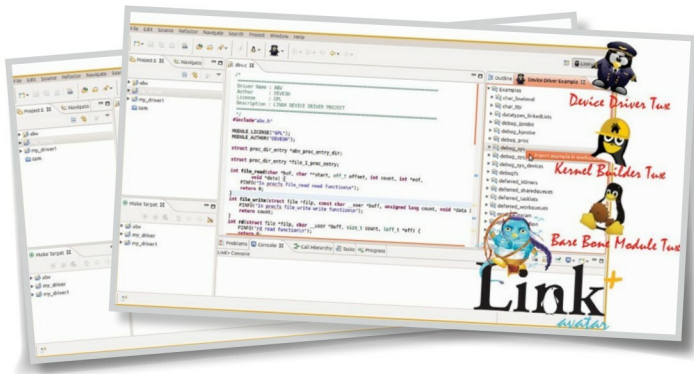


The Link+ Project for Linux Kernel Developers

Kernel developers will be interested to learn how to use the LinK+ IDE, which has been developed by a team from C-DAC. The article is detailed and comprehensive.



In this article, discover how Linux kernel programming can be simplified using an IDE. Let's take a look at the IDE named LinK+ which supports Linux kernel configuration, compilation and emulation, system call development and device driver development. At the end of this article, developers of Linux device drivers are sure to start loving it!

The Link+ IDE is a simple IDE for Linux kernel developers. It is based on the Eclipse IDE and is customised for Linux kernel programming. It reduces development time and executes code in an elegant fashion. The Link+ IDE includes various templates in the categories of character, block and network device driver sub-systems for device driver development. It includes various bus infrastructure templates and also provides Linux device driver code completion, code assistance and code navigation for developers.

Features of the LinK+ IDE

The first release of this IDE is called *LinK+ Avatar* and it supports the following set of features:

- Hardware architectures: x86 and x86_64

- Linux device driver auto code generation, completion and navigation
- Wizard for kernel configuration and compilation
- Reduced kernel compilation time through *ccache* and *distcc*
- QEMU emulator support for Linux kernel image emulation and debugging
- Linux system call development
- Supports both Debian and RPM Linux families
- Supports Sparse for static analysis of Linux device drivers

LinK+ IDE installation

Prerequisites for Ubuntu Linux distributions

Before installing LinK+ IDE, the following set of packages needs to be installed on your machine. You can run the command shown below:

```
dileep@dileepk:~$ sudo apt-get install openjdk-6-jdk libqt4-  
dev libcurses5 sparse qemu qemu-system openssh-server  
openssh-client bridge-utils iptables dnsmasq linux-headers-  
`uname -r`
```